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METACOMPREHENSION DURING RARE WORD COMPREHENSION

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To examine metacomprehension during comprehension, undergraduates ($n = 133$) were asked to provide descriptions of how they determined the meaning of four rare words presented in short passages. Content analysis of these written descriptions revealed task-specific metacomprehension reflecting lexical, textbase, and situation model processes. Cluster analysis yielded four metacomprehension groups with differing emphases on textbase and situation model processes. Participants also provided definitions, post-comprehension assessments, and completed the Metacomprehension Scale. Differences across the subgroups provide evidence that individual differences in task-specific metacomprehension influence interpretation accuracy and post-comprehension assessments. In contrast, global metacomprehension subscales were uncorrelated with interpretation accuracy.

Introduction

An individual's metacognitive notions may include (a) ideas about how cognitive processes function in general (metacognitive knowledge); and (b) ideas about how *well* particular cognitive processes have functioned in the past or will function in the future (Flavell, 1979; Hacker, 1998a; Schraw & Moshman, 1995). Metacognition research has emphasized a variety of metacognitive activities (e.g., feelings, experiences, strategies) across a variety of cognitive processes (e.g., memory, comprehension, and problem-solving). Given that metacognitive *activities* and *thoughts* may be specific to the cognitive task an individual is engaged in, analyses of task-specific processes has the potential to augment this literature. In addition, the nature of these thoughts and activities may reflect metacognitive *ability* as it pertains to that task.

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Hence, metacognitive activity may be domain-specific with respect to (a) the nature of the metacognitive activity; and (b) an individual's ability in that particular domain. In addition, task-specific metacognitive activities and abilities may influence processing and processing outcomes as much as global metacognition.

Metacomprehension is a specific type of metacognition, referring to ideas people have about their own comprehension processes and about comprehension in general. Empirical endeavors pertaining to metacomprehension have included studies examining the accuracy of readers' predictions of comprehension (Maki & Berry, 1984; Maki, Jonas, & Kallod, 1994; Moore, Lin-Agler, & Zabrocky, 2005; Weaver, 1990); the relationship between global metacomprehension and performance (Lin, Moore, & Zabrocky, 2000; Moore, Zabrocky, & Commander, 1997a, 1997b; Weaver & Bryant, 1995); scientific text comprehension (Dunlosky, Rawson, & Hacker, 2002); and metacomprehension in children (Cataldo & Cornoldi, 1998; Lovett & Pillow, 1995; Schneider & Pressley, 1989). To extend the scope of the literature on metacomprehension, the present study addresses the nature of metacognitive activity as it pertains to a rare word comprehension task in adults.

Theoretical and empirical endeavors related to metacomprehension have been influenced by the general metacognition literature. At a global level, two general types of metacognitive processes have been proposed: processes related to *monitoring* and processes related to *control* (Hacker, 1998b; Nelson & Narens, 1990). In the domain of text comprehension, monitoring processes pertain to *monitoring* for discourse inconsistencies or comprehension problems (Glenberg, Wilkinson, & Epstein, 1982). In other words, readers are thought to engage in monitoring for comprehension challenges that originate in the text (e.g., its coherence) or in the readers' own processes. When comprehension challenges arise, readers engage in process regulation (control) in an effort to address these effectively (Flavell, 1979; Nelson, 1996; Nelson & Narens, 1990). For example, most readers examine emerging meanings for consistency with meanings derived previously, thereby monitoring comprehension. If a discrepancy is noticed, then readers may select to reread previously read sentences or paragraphs for misinterpreted information, thereby regulating understanding processes.

Theoretical and empirical endeavors related to metacomprehension have also been influenced by theoretical models of discourse comprehension, although the latter have only recently been emphasized (Dunlosky et al., 2002). Models of discourse comprehension describe comprehension as involving three levels of representation: the lexical level, the textbase level, and the mental or situation model (Glenberg et al., 1982; Graesser, 1981; Graesser, Millis, & Zwaan, 1997; van Dijk & Kintsch, 1983). Mental representations at these three levels differ as follows: (a) at the lexical level, exact words and syntax are represented verbatim; (b) at the textbase level, the meaning of the explicit text is represented in propositional form involving integration of explicitly stated ideas; and (c) at the situation model level, what the text corresponds to in the real world is mentally represented.

The lexical level pertains to the mental representation for each explicitly stated word and its meaning. The textbase level goes beyond the single word and its meaning to ideas that provide a basic understanding. The development of textbase mental representations includes integrating across basic ideas to develop a coherent sequence of events for narrative texts or coherent connections among ideas for expository texts. In short, readers are able to develop coherence at a local level after developing textbase mental representations, with the result being a basic understanding of the text. Situation model processes include generating relevant inferences using both lexical and textbase representations. For narrative texts, situation models reflect what could have (or has) occurred in the real world. In other words, situation models are mental representations of the events if they had actually occurred in the real world.

Interpreting a rare word encountered in a supportive passage should lead to the examination of mental representations at all three levels of representation. Naturally, skilled adult readers begin at the lexical level, searching for words related to a rare word's meaning (verbatim meaning-relevant cues). One or two words and their meanings are typically not sufficient, so readers begin combining ideas to generate propositions linking explicitly stated ideas. And finally, developing a reasonable interpretation necessitates mentally accessing the situation model (the third level), as well as any meaning-relevant information, that can be inferred from that model. This is essential when rare words pertain to the

central problem in each story. Hence, developing a contextually relevant interpretation of each rare word depends on the mental representations at all three levels, with the quality of these mental representations influencing the accuracy and specificity of the interpretation.

In this study, participants read passages with rare words and were asked to provide written narratives describing their interpretation processes, referred to as *process narratives*. Content analysis of these process narratives was designed specifically to examine metacognitive activity as it pertains to text comprehension processes. More specifically, we coded the narratives for inclusion of verbatim cues, integration of cues, and for meaning-relevant inferences reflecting three levels of mental representations: lexical, textbase, and situation model, respectively.

We hypothesized that: (a) the majority of the narratives would include verbatim cues, evidencing lexical level processes; (b) fewer narratives would include cue integration, evidencing textbase processes, and/or meaning-relevant inferences, evidencing situation model processes; and (c) process narratives would reflect different configurations of metacognitive activity associated with individual differences in the metacomprehension activities and abilities. Specifically, we expected some participants to have high scores across all three levels (lexical, textbase, and situation model), reflecting more awareness of comprehension processes (metacomprehension). In addition, we predicted that these participants would have interpretation and confidence scores that reflected a higher level of accuracy. We also expected that some participants would have other configurations of metacognitive activity, (e.g., high levels of verbatim cues and integration but low levels of inferential activity) and that interpretation accuracy and confidence in these participants would be reduced compared to those high across all three levels of metacomprehension activity.

The advantage of using the rare word comprehension task rather than an overall comprehension task is that it is possible to focus on the nature of each participant's interpretation with more specificity. For example, the verbatim cues and inferences relevant to interpreting a particular rare word were identifiable for the purposes of content analysis. As such, the use of this approach

enhanced the precision of the coding process. In previous studies, unfamiliar word comprehension has been used to examine the role of working memory and comprehension (Daneman & Green, 1986) and to examine late-life changes in text comprehension (McGinnis & Zelinski, 2000, 2003).

Global metacomprehension should also affect performance across a variety of comprehension tasks, perhaps as much as or more than task-specific metacognitive activities. Empirical endeavors related to global metacognitive beliefs and dispositions reflect individual differences in global metacognition and the role of these differences in comprehension and judgments of comprehension (Lin et al., 2000; Moore et al., 1997a, 1997b). For example, an individual's global metacognitive notions could include an awareness that careful and thorough processing of a difficult text may improve comprehension. This idea constitutes a metacomprehension belief and disposition that would apply in many scenarios (reading a novel, a journal article, or short texts, like instructions). Alternatively, individuals who are less aware of the extra processing difficult texts necessitate should comprehend less thoroughly and accurately. Hence, this awareness could be thought of as a *global* metacomprehension belief.

After reading all four passages, producing definitions and process narratives, participants responded to the Metacomprehension Scale (Moore et al., 1997a), an instrument designed to assess various global ideas about metacomprehension. It was hypothesized that the correlations among global metacomprehension subscales and performance indicators would resemble correlations among task-specific metacomprehension scores and performance.

In summary, this study was designed to examine: (a) configurations of metacognitive activity across individuals, particularly as these pertain to varying degrees of emphasis on lexical, textbase, and situation model processes; (b) the effect of metacognitive processing configurations on interpretation accuracy and on confidence ratings; and (c) the magnitude of correlations among interpretation accuracy and metacomprehension, with an interest in comparing the correlations obtained for global metacomprehension and interpretation variables to the correlations obtained for task-specific metacomprehension and interpretation variables.

Method

Participants

Participants consisted of 133 undergraduates (41 males, 92 females) from a large urban university California State University, Long Beach. Participants were recruited from a variety of general education psychology courses and received extra credit for their participation. The average age was 21.83 (range 18 to 58 years of age). All participants were either native English speakers ($n = 111$) or were nonnative speakers who attended English-only schools since kindergarten ($n = 22$). On average, educational attainment was 1.48 years of college ($SD = 0.93$) for a total of 13.48 years of education.

Materials

VOCABULARY

Participants provided written definitions for the second half of the Wechsler Adult Intelligence Scale (WAIS-R) Vocabulary subtest (Wechsler, 1981). Responses were scored according to the WAIS-R scoring guidelines. $M = 19.83$, $SD = 6.46$.

RARE WORD STORIES

Four passages used in previous studies of rare word comprehension (Daneman & Green, 1986; McGinnis & Zelinski, 2000, 2003) were utilized. Each passage consists of around 150 words describing a protagonist dealing with a problem. Each passage includes a rare word, the meaning of which is associated with the main idea of the passage. None of the 133 participants reported encountering any of these rare words prior to participation. The passage for *dippoldism* is provided below.

Johnson had held his present job since 1989 but in the past year there had been a number of complaints. After his superiors investigated, Johnson was fired for *dippoldism*. When his superiors found out what Johnson had been doing they were quite disturbed. In some cases, the welts had taken a long time to heal. Many of the victims had broken rules and disciplining them was within Johnson's duties, but his method had injured some of them. This practice, although once common, was no longer considered acceptable. Even though most of the victims had been scared to complain, eventually

the story came out and Johnson was fired. Johnson's replacement emphasized learning rather than discipline and there was satisfaction among the parents with the decision.

JUDGMENT OF COMPREHENSION (CONFIDENCE RATINGS)

After reading each story and generating a definition, participants were asked to rate how confident they were in the accuracy of their definition (1 = *not at all confident* to 8 = *very confident*). Added together, these confidence ratings comprise an individual's overall confidence score and reflect an individual's post-comprehension assessment.

PROCESS NARRATIVES

After defining the word and rating their confidence, participants were asked to write a *process narrative* for each passage. The instructions for the passage containing the word *dippoldism* read as follows: "Please describe how you figured out the meaning of the word dippoldism. This information is very important for this investigation, so please feel free to use whatever kinds of language that you like to tell us about this process."

METACOMPREHENSION SCALE

To examine global dispositions about metacomprehension, a revised version of the Metacomprehension Scale (MCS) was administered (Moore et al., 1997a). This instrument includes 22 statements about various aspects of metacomprehension and asks participants to rate their agreement on a 6-point scale: 1 (*strongly disagree*) to 6 (*strongly agree*). Ten statements from an unpublished inventory previously used in our lab were added. Principle components factor analysis yielded seven factors with eigenvalues over 1.00 and these factors accounted for 66% of the variance. Statements that did not yield factor loadings of at least .45 were eliminated, resulting in a loss of 10 statements.

Six metacomprehension factors obtained in our data were identical to or conceptually consistent with the MCS subscales: anxiety, confidence (referred to as capacity in Moore et al., 1997a), strategy use, valuing comprehension skill, effort, and task characteristics. A new factor, assessing skill in others, was obtained from the statements added.

PROCEDURE

Participants were tested in groups of three, four, or five. They were given a booklet that contained instructions, the four rare word passages, the metacomprehension questionnaire, the vocabulary test, and a background information sheet. After reading the instructions, participants were asked to read the first passage, to provide a definition for the rare word therein, to rate their confidence in the accuracy of their definition, and to complete a process narrative. This sequence was repeated for the remaining three passages. Then participants responded to the questions in the Metacomprehension Scale, completed the vocabulary test, and the background information sheet. Passage order was varied to address fatigue confounds.

CONTENT ANALYSIS

All background sheets were removed before coding definitions and process narratives so that all five coders were blind to the age, sex, education, and vocabulary score. Two coders coded each definition and each process narrative. Coding disagreements were discussed until consensus was achieved. Coding disagreements usually involved one coder completely missing a word or a phrase rather than conceptual disagreements. Non-consensus events were rare (less than 2% overall) and were resolved by discussion with a third coder.

CONTENT ANALYSIS—RARE WORD DEFINITIONS

The code used for coding definitions consisted of the three meaning components associated with each word with each meaning component coded as present (1) or absent (0). The three meaning components for *dippoldism* are beating, children, and school. For example, the definition *beating children* was scored as one for beating and one for children. The definition *corporal punishment* was scored as one because it strongly implies beating. After coding was complete, meaning component totals were computed. Internal consistency for meaning component total was assessed using a split-half (odd/even) approach with the Spearman-Brown adjustment to estimate the reliability under non-split conditions, yielding a moderate level of internal consistency, $r = .646$. Definitions were also coded for being completely off-target,

yielding total off-target scores for each participant ranging from 0 to 4.

CONTENT ANALYSIS—PROCESS NARRATIVES

Content analysis of the process narratives yielded three metacomprehension scores: cue selection, integration, and inference, reflecting lexical, textbase, and situation model processes as described previously. For example, a person determining the meaning for the word *dippoldism* could state that they used the idea (cue) of *welts* when determining its meaning. The six most commonly mentioned cues for each of the four passages were used when coding for cues selected. A cue selection score of 1 was assigned when a process narrative included at least one of the six cues or a 0 if the narrative did not include any of the six cues.

Process narratives were also coded for cue integration (e.g., “I combined these ideas” or if several cues were used in the same sentence). Because cue integration reflects integrating across ideas to develop a propositional structure, this score reflects a reader’s metacognitive awareness of textbase processes.

Process narratives were also scored for inference generation related to situation model processes. If a participant generated an inference relevant to the rare word’s meaning, they received an inference score of 1 for that passage. For each of these variables, each participant’s 0–4 score was converted to percentages.

Participants were also coded a 1 if they reported rereading the passage for more information and a 0 if they did not. Rereading totals for each participant were computed.

Results

DEFINITION ACCURACY, CONFIDENCE, AND METACOMPREHENSION VARIABLES

On average, participants produced just under one meaning component for each passage, $M = 0.87$, $SD = 0.42$, indicating that low scores were common (0–1 out of a possible 3 per passage). This score reflects the tendency for most participants to generate an interpretation that included the most central idea of a word’s meaning (for a score of 1 meaning component out of 3 possible) and that generating the precise

meaning was less common (a score of 3 out of 3). In addition, participants were somewhat unlikely to produce off-target responses, $M = 0.76, SD = 0.85$, revealing that they scored low on off-target responses (0 or 1 were most common out of a possible total of 4 off-target responses). These results are consistent with previous research using rare word comprehension methodology (McGinnis & Zelinski, 2000).

Overall, participants' confidence ratings were above average, as evidenced by a confidence rating mean of 5.03 ($SD = 1.03$) on a 7-point scale.

The metacomprehension variable selection was associated with the largest percentage, $M = 86.84\%, SD = 21.23\%$, reflecting the presence of at least one verbatim or lexical cue in almost every process narrative. In contrast, the averages for the metacomprehension variables integration and inference were substantially lower: $M = 33.83\%, SD = 29.23\%$; and $M = 32.14\%, SD = 29.29\%$, respectively.

Only 3 of the 133 participants (just over 1%) mentioned rereading the passage in their process narratives, suggesting that participants were not inclined to reread passages. Because of the low frequency of rereading, this variable is not included in subsequent analyses.

METACOMPREHENSION CORRELATES OF DEFINITION ACCURACY

Correlations among the metacomprehension variables (selection, integration, and inference) and definition accuracy (meaning component and off-target scores) were obtained and are provided in Table 1. Meaning component was correlated significantly and positively with all three metacomprehension variables, suggesting that participants who had higher levels of metacomprehension awareness produced better definitions. Meaning component was also positively correlated with confidence, indicating that participants who rated confidence high also produced more meaning components for each rare word. Off-target was correlated negatively and significantly with selection, inference, and confidence scores.

Correlations between vocabulary and metacomprehension variables and between WAIS-R Vocabulary and Metacomprehension Scale (MCS) subscale scores are presented in the last column of Table 1. Correlations between MCS subscales and definition

TABLE 1 Correlations Among Definition Accuracy, Vocabulary, and Metacomprehension Variables

	MC	Off	Voc
Task-Specific Variables			
Selection	.234**	-.220**	.002
Integration	.220**	-.082	.166
Inference	.224**	-.274***	.300***
Confidence Total	.277***	-.235**	.344***
Global Variables (MCS)			
Anxiety	-.143	.160	-.208**
Confidence	.091	-.192*	.344***
Value Comp	-.020	-.008	.139
Personal Strategy	-.037	-.126	.035
Assessing Others	.018	-.020	.027
Dispositions—Effort	-.024	.011	.082
Task Characteristics	-.176*	.174*	.345***

Note. Pearson-product moment correlations for $n = 133$; MC = meaning components; Off = off-target; Voc = WAIS-R Vocabulary.

* $p \leq .05$; ** $p \leq .01$; *** $p \leq .001$.

accuracy variables were weaker than expected, and most of these were nonsignificant. However, it is interesting to note that MCS subscales were more likely to be correlated with vocabulary than with rare word definition accuracy.

CLUSTER ANALYSIS OF TASK-SPECIFIC METACOMPREHENSION VARIABLES

Cluster analysis was used to group participants according to metacomprehension score configurations. Quick Cluster in SPSS, a nonhierarchical procedure using a parallel threshold method, yielded a four-cluster solution that was conceptually sound and virtually identical to the four-cluster solution obtained using Ward's hierarchical procedure. The solution obtained for integration and inference was identical to the solution obtained for all three metacomprehension variables, indicating that the variable selection was not discriminating among participants, an outcome that probably reflects ceiling effects for this variable. In addition, the 10 most discrepant cases (participants with the largest cluster center deviations) were removed and the cluster analysis was rerun. Removing these discrepant cases had no impact on the solution, so all participants were included in the final analysis.

Cluster center scores for cluster 1 were low on both integration and inference (13.27%, 11.22%). Cluster 2 center scores were high on integration but low on inference (65.52%, 10.34%). Cluster 3 was associated with low integration but very high inference scores (13.28%, 65.63%), whereas cluster 4 center scores were high for both integration and inference (66.30%, 57.61%). The low-low group consisted of the most participants ($n = 49$), followed by low-high ($n = 32$), high-low ($n = 29$), and high-high ($n = 23$). Univariate analysis of variance was used to validate the clustered subgroups with subgroup metacomprehension differences supporting the integrity of this cluster solution. These results are provided in Table 2.

This analysis reveals that participants can be categorized meaningfully according to their awareness of (a) textbase processes (integration) and (b) situation model processes

TABLE 2 Comprehension and Metacomprehension Statistics for Metacomprehension Subgroup Analyses

Metacomprehension Groups										
	Low-Low (n = 49)		High-Low (n = 29)		Low-High (n = 32)		High-High (n = 23)		ANOVA	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>F</i> (3, 129)	η^2
Metacomprehension Variables										
Sel	75.00	27.48	97.41	10.23	89.84	14.00	94.57	10.54	10.61***	.198
Int	13.27	12.61	65.52	16.92	13.28	12.68	66.30	14.32	149.87***	.777
Inf	11.22	12.56	10.34	12.53	65.63	17.68	57.61	15.87	135.95***	.760
Comprehension Variables										
Voc	18.04 _a	5.56	19.17 _b	5.76	20.25	6.77	23.91 _{ab}	7.12	4.845**	.101
MC	0.55 _a	0.37	0.95	0.37	0.88	0.38	1.07 _a	0.53	4.263**	.090
Off	1.04	0.98	0.62	0.78	0.59	0.61	0.57	0.79	3.018*	.066

Note. Low-low = low integration–low inference; High-low = high integration–low inference; Low-high = low integration–high inference; and High-high = high integration–high inference. Sel = selection; Int = integration; Inf = inference; Voc = WAIS-R Vocabulary (max = 40); MC = meaning-component (max = 3); Off = off-target (max = 4). Metacomprehension scores reflect the average percentage of the four passages containing at least one instance of selection, integration, and inferencing. The quantity in the first cell, 75.00, corresponds to an average of 75% or three quarters of each participant's process narratives. Comprehension means with the same subscripts differed significantly at $p < .05$.

* $p \leq .05$; ** $p \leq .01$; *** $p \leq .001$.

(inferencing). Participants were either low integration–low inference (low-low), high integration–low inference (high-low), low integration–high inference (low-high), or high integration–high inference (high-high), reflecting different configurations of metacomprehension across participants.

DEFINITION ACCURACY ACROSS METACOMPREHENSION SUBGROUPS

Univariate analysis of variance procedures were used to examine the differences among metacomprehension subgroups. Means, standard deviations, and ANOVA statistics for these comparisons are provided in Table 2. Tukey's HSD procedure was used to examine post hoc mean comparisons for vocabulary, meaning component, and off-target scores with significant differences indicated with subscripts.

CONFIDENCE AND COMPREHENSION ACCURACY

Correlations for confidence ratings and definition accuracy within each metacomprehension group were obtained. The correlation for confidence ratings and off-target definitions were significant in the high-high group, $r(21) = -.467$, $p < .05$, suggesting that this group was more likely to be less confident when generating off-target definitions. The correlations for confidence ratings and off-target scores were nonsignificant in the other three groups, suggesting less sensitivity to interpretation accuracy.

Significant correlations for confidence and meaning component scores were obtained for 3 of the 4 groups: low-low, $r(47) = .444$, $p < .01$; high-high: $r(21) = .620$, $p < .01$; and a negative correlation for the low-high group, $r(30) = -.414$, $p < .05$. The significant *positive* correlations indicate that confidence accuracy pertained to participants who were low in both integration and inference as well as those who were high in both. These results reveal that many participants accurately evaluated the integrity of their processes, even when these processes were less effective. In contrast, the *negative* correlation for the low-high group reflects overconfidence in definition accuracy. In this group, participants were high in inference (situation model) but low in integration (textbase) scores. Hence, overconfidence was greatest in participants whose process narratives reflected an emphasis on inferential processing but a deemphasis on textbase processes. In other words, these participants were using a strategy that resulted in an

interpretation that seemed reasonable to them because of their own confidence in their approach. Taken together, these results suggest that the nature of an individual's metacomprehension processing models affect post-performance assessments as well as performance.

Discussion

TASK-SPECIFIC METACOMPREHENSION AWARENESS

One of the goals of this study was to examine the nature of task-specific metacomprehension during text comprehension using a rare word approach. As hypothesized, process narratives (participant-generated descriptions of their comprehension processes) revealed that participants emphasized lexical-level processes (87% of the process narratives included at least one verbatim cue). These findings are consistent with our expectation that process narratives would reflect lexical processes because these are fundamental to the comprehension process. In comparison, only about one third of the process narratives included evidence of textbase and situation model processes (34 and 32%, respectively).

Participant differences in task-specific metacomprehension variables resulted in four metacomprehension groups, with varying degrees of emphasis on textbase (integration) and situation model (inference) processes. As hypothesized, the differences in interpretation accuracy (definition accuracy) across these groups were significant, with the highest accuracy scores attained by the high-high group (high integration–high inference) and the lowest by the low-low group (low integration–low inference). These results provide evidence that awareness of task-specific metacognitive activity plays a role in comprehension accuracy. The most surprising metacomprehension subgroup was the low-high group (low integration–high inference). Their process narratives reflected high levels of situation model processing even though textbase scores were low. In short, this group appeared to skip a basic comprehension process (cue integration) even though they generated meaning-relevant inferences.

The association between interpretation accuracy and post-comprehension assessments (confidence ratings) also differed across the metacomprehension subgroups. The post-comprehension assessments for the high-high group were remarkably

accurate; in other words, their level of confidence in the accuracy of their individual definitions was related to actual accuracy, yielding a significant positive correlation, $r = .620$, $p < .01$. The low-low group were also relatively accurate, $r = .444$, $p < .01$, reflecting sensitivity to lower levels of interpretation accuracy. These results suggest that many comprehenders assess immediate performance with some level of accuracy.

However, the negative correlation, $r = -.414$, $p < .05$, for the low integration–high inference group suggests a level of overconfidence that is unwarranted given the diminished accuracy of their interpretations: results that were not expected. These comprehenders seem prone to overestimating the accuracy of their interpretations and present an interesting contrast to the high-high and low-low participants, who were more accurate in their estimates. Because these participants deemphasized integrative processes, their mental representations may have lacked the specificity necessary for developing a reasonably accurate interpretation. This group also had low selection scores, reflecting a tendency to deemphasize lexical level processes as well. The overconfidence exhibited by individuals in this subgroup may reflect a metacognitive awareness of inferential processing, possibly eliciting the sense that this sophisticated approach should result in an accurate interpretation, even though it did not.

GLOBAL METACOMPREHENSION

The almost nonexistent relationship among global metacomprehension and interpretation accuracy was not expected: none of the MCS scores, indicators of global metacomprehension, were correlated with meaning component scores. The relationship between global metacomprehension and performance may be a function of the extent to which rare word interpretation depends on task-specific processing approaches and strategies rather than on global approaches. Hence, global metacomprehension may not predict performance on a rare word comprehension task as reliably as task-specific metacomprehension scores (selection, integration, and inference), which *were* correlated with meaning component scores. More specifically, accurate meaning attainment in this task may be less dependent on a comprehender's global dispositions than on task-specific metacognitive awarenesses.

It is also noteworthy that three of the MCS subscales were correlated with vocabulary scores (anxiety about comprehension, confidence in comprehension ability, and sensitivity to task characteristics). Because vocabulary scores reflect comprehension proficiency across a variety of settings, vocabulary may provide a global measure of comprehension proficiency, and, as a result, may more be more likely to reflect the effects of global metacognitive dispositions. Hence, the data herein suggest that global metacomprehension may play a minor role in specific comprehension tasks, an idea that is consistent with evidence that metacognitive ability may be task-specific rather than general (Kelemen, Frost, & Weaver, 2000).

Moore et al. (2005) recently examined the hypothesis that accuracy self-evaluations may be less likely to reflect an individual's assessment of a task just performed, and more likely to reflect general assessments of past experiences in a particular cognitive domain. Given this, global measures of metacomprehension should be as correlated with task performance as immediate assessments. In our study, task-relevant metacomprehension and post-comprehension assessments were related to performance (definition accuracy), but global metacomprehension scores, reflecting general dispositions about comprehension, were uncorrelated with almost every aspect of current task performance.

The data herein suggest a second possibility with respect to post-comprehension assessments: that individual differences in metacognitive activities and abilities have differing influences on performance and performance assessments. As hypothesized, the group with the highest metacognitive scores produced the most accurate definitions and the most accurate post-comprehension assessments. Similarly, the group with the lowest metacognitive scores also assessed their comprehension reasonably well, even though their interpretations were less accurate. In contrast, the group who overrated their performance included those who demonstrated less metacognitive awareness of lexical and textbase processes (the simplest and most obvious processes), even though they demonstrated awareness of situation model processes. In short, our data reflect what metacognitive theorists have long thought about metacognition: that metacognitive activities and dispositions vary across individuals, resulting in assessment differences as well as performance differences on cognitive tasks.

Unlike most studies in this literature, the present one used a language processing model as a guide in identifying three kinds of task-relevant metacognitive activity, enabling an examination of individual differences across these kinds of metacognitive activities as well as the role that these individual differences have on performance and on performance assessments. The use of a rare word comprehension task and the use of process narratives enabled this kind of examination. Our results suggest that metacognitive activity and its influence on performance and on performance assessments may be more complex than the literature suggests, with individual differences influencing performance in various ways.

USING PROCESS NARRATIVES TO EXAMINE COMPREHENSION PROCESSES

In the present study, participant-generated process narratives reflected the text processing approaches used to comprehend rare words presented in short passages. Methodologically, this study provides evidence that process narratives may allow researchers to assess certain types of metacognitive processes. Even though a limitation of this approach could be that narrative quality depends on an individual's ability to articulate mental activity with the possibility that these abilities could be lacking even when comprehension is proficient, our data appear to refute this limitation. If this were a serious limitation, then some would comprehend well even though they produced impoverished narratives. This pattern, impoverished narratives in light of adequate performance, should result in weak relationships among task-specific metacomprehension and performance. But that was not observed: the relationships among these variables were fairly robust. In short, this study suggests that process narratives and content analysis of these narratives may be a useful methodology for studying metacognitive activity.

EDUCATIONAL IMPLICATIONS

Our findings suggest that many college-aged individuals may be aware that selecting and integrating appropriate meaning-relevant cues improves comprehension during some types of tasks. Even though there was less awareness of integrative and inferential processes, this type of awareness was present in about one third of the process narratives. Most noteworthy, however, is

the subgroup whose process narratives included fewer instances of explicitly stated meaning-relevant information (selection and integration scores), even though their narratives suggested awareness of implicit aspects of comprehension related to situation model processes. The possibility exists that these individuals comprehend rapidly, moving beyond verbatim representations quickly, a possibility that would be associated with interpretation scores that, at the very least, resembled participants with higher cue integration scores. As this similarity was not observed statistically, the idea that efficient processing characterizes this group is unsupported.

The findings herein suggest that the tendency to emphasize inferential processes while deemphasizing selection and integration comprehension processes may result in interpretations that range from overgeneralized to grossly inaccurate. College-aged individuals, most of whom encounter challenging comprehension tasks frequently, may experience frustration or even confusion when they find that their text interpretations are inaccurate, particularly if they presume that their approach is sophisticated compared to other individuals or compared to approaches they used at younger ages.

It is possible that the individuals in this subgroup are in the midst of a developmental transition to the kinds of comprehension processes that characterize mature adult comprehenders, with this phase associated with overuse of the new approach, similar to children's overuse of certain grammatical and syntactical forms during language acquisition (Brown, 1968, 1973). However feasible this may seem, this idea is speculative given the lack of empirical support for such a transition in the reading and comprehension literature.

From an educational perspective, it may behoove instructors and professors to watch for college students exhibiting this approach, possibly encouraging them to attend to verbatim information as well as to the inferential or more implicit aspects relevant to a comprehension task. It might even be useful educationally to incorporate process narratives in courses focused on improving reading comprehension in an effort to identify this type of problem, providing a tool to demonstrate to individuals exhibiting this pattern how they could improve text comprehension with some specificity.

Conclusions

This study suggests that task-specific metacognitive activity plays an important role in comprehending rare words presented in contextually supportive passages. In addition, subtypes of metacognitive activity in individuals may be more nuanced than metacognitive researchers had previously imagined, possibly revealing a pattern that is counterintuitive. Empirically identifying (a) task-specific metacognitive variables and (b) configurations of metacognitive processing subgroups across a variety of cognitive tasks has the potential to yield results that are both educationally relevant and theoretically fruitful.

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